

<https://www.ti.com/lit/ds/symlink/opa818.pdf>
p27

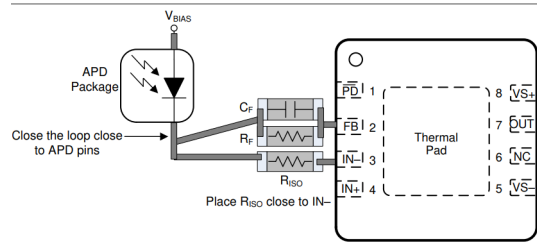


Figure 60. Improved TIA Layout

Photodiodes with TO-18 footprint:

Farnell 1495575
Hamamatsu S5973
D= 400um active area
Cd=1.6 pF @ Vr=3V3

Thorlabs FDS015
D=150 um active area
0.65pF @ Vr = 5 V

Thorlabs FDS025
D= 250 um active area, with ball lens
0.94 pF @ Vr = 5 V

Hamamatsu S5972
D= 800um active area
3 pF @ Vr = 10 V

Hamamatsu S5971
D= 1.2 mm active area
3 pF @ Vr = 10 V

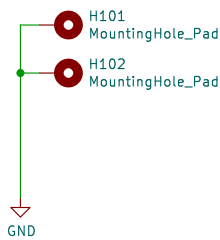
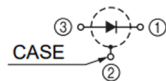
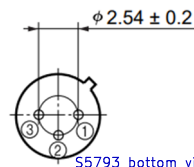
OSI FCI-125G-006HRL
Hamamatsu S9055(-01)

Thorlabs FGA01FC, InGaAs, 2pF @ 5V
same as OSI INGAAS-120L-FC

Thorlabs FGA015, InGaAs, 1.5 pF @ 5V
D = 0.15 mm active area

FGA01, ball lens

For estimating Bandwidth, Transimpedance-amplifier noise, shot-noise limit, etc. see:
TIASim at <https://github.com/aewallin/TIASim>



PSU:
Thorlabs LDS12B
±12 VDC Regulated Linear Power Supply, 6 W
Lumberg RSMV3 male connector, 3-pin

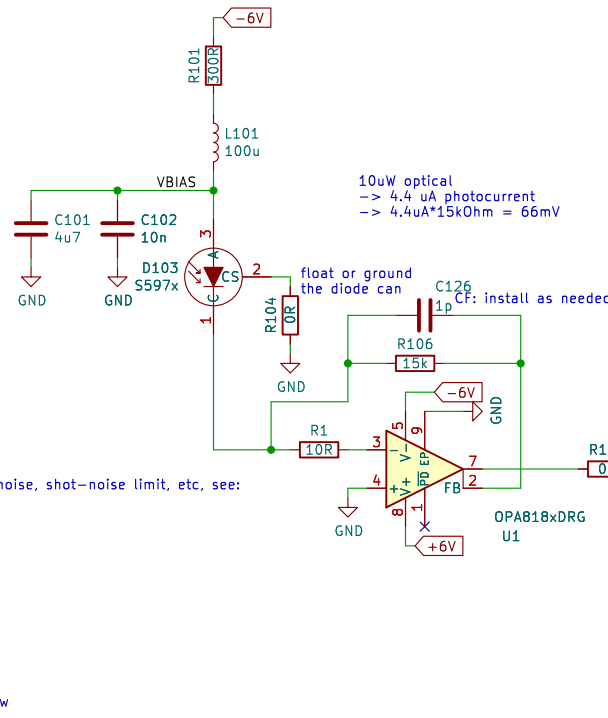
PSU-lead: Digikiy A120947-ND (mates with Thorlabs 3-pin psu connector)
3-pin M8 female connector
1.5 m 3-wire lead, solder to PCB

See MeasurementData sheet for measurements and
noise-models for built 2-stage detectors:

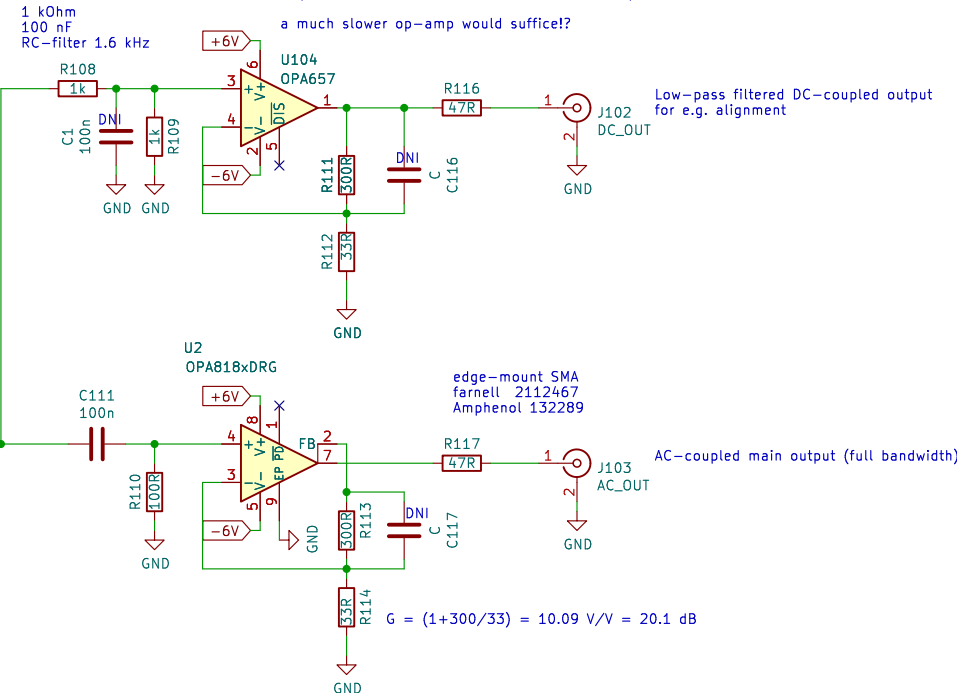
MeasurementData



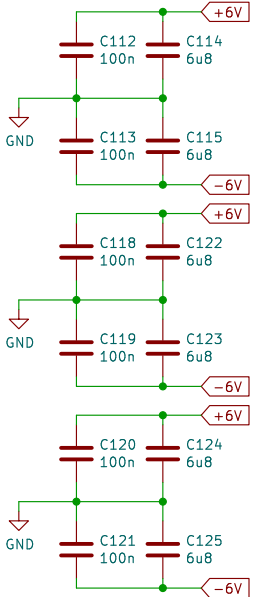
File: 2stageTIA_measurementData.kicad_sch



use OPA657 here?
<https://www.ti.com/lit/ds/sbos197f/sbos197f.pdf>
a much slower op-amp would suffice!?



Decoupling caps for three op-amps



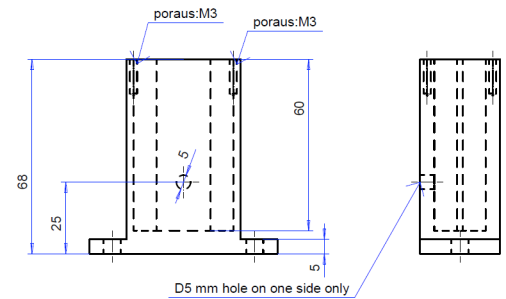
$$G = (1 + 300/33) = 10.09 \text{ V/V} = 20.1 \text{ dB}$$

2019 detector (OPA657) gain:
78 dBohms transimpedance (8k2 ohms)
+20 dB postgain (10V/V)
+28 dB postgain (ZFL-500LN)
126 dB total

2019Modified detector (OPA657) gain:
92 dBohms transimpedance (39k ohms)
+20 dB postgain (10V/V)
112 dB total

2022 detector with OPA818
90 dBohms transimpedance (33kOhm)
20 dB postgain (10 V/V)
110 dB total

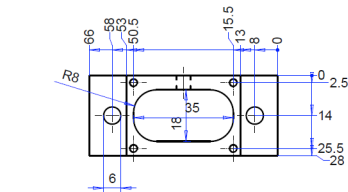
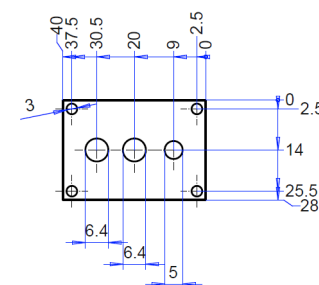
Mechanical design of Aluminium enclosure
Optical beam-height over table 25mm
Mounting-holes for M6 screws on 25mm grid



Mechanical design for enclosure top



D6.4 mm holes for SMA-connectors
D5 mm hole for power-supply cables



2022-10 built with FGA015 InGaAs photodiode and Rf=30kOhm, ca 44 MHz BW
2022-06: built 1st board with Rf=33kOhm, 10 V/V postgain - measurements look good.
anders.e.e.wallin "at" gmail.com

Sheet: /
File: working.kicad_sch

Title: 2-stage OPA818 photodetector

Size: A3
KiCad E.D.A. eeschema 7.0.6

Date: 2019-12

Rev: draft

Id: 1/2

1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

Sheet: /MeasurementData/ File: 2stageTIA_measurementData.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. eeschema 7.0.6	Id: 2/2	